

RISHIKESH PATIL

+91-77188 66002 | rishikeshpatil0605@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

SUMMARY

Applied AI/GenAI Engineer | RAG Systems | Agent Orchestration | NLP | AI Engineering

Final-year B.Tech student in Electronics & Computer Science (**CGPA: 9.65/10, Top 5 in Department**) with hands-on experience building AI, ML, and full-stack systems across RAG, agentic workflows, speech emotion recognition, and continual learning. Research author with work targeting **Springer-indexed venues**, including AR-EWC and ACE-SER. Currently an SDE Intern at AyriTech and previously at Tata Consulting Engineers and led a **20+ member IEEE technical team**.

EDUCATION

Vidyalankar Institute of Technology

B.Tech in Electronics and Computer Science (Final Year); **CGPA: 9.65/10 | Rank: Top 5 in Department**

Mumbai, Maharashtra

Aug 2023 – Present

DAV School

10th Board (S.S.C): 89.4% | 12th Board (H.S.C): 88%

Mumbai, Maharashtra

2023

RESEARCH PAPERS

Adaptive Recursive Elastic Weight Consolidation for Continual Learning in ANFIS-Based Building Energy Prediction | First Author

2026

- 3rd Int'l Conf. on Information Technology & Intelligence (ITI 2026), Springer LNNS – Scopus indexed.
- Introduced **AR-EWC**, a constant cost, recursively re-anchored penalty for continual ANFIS updates, benchmarked against **9 baselines across 5 seeds** on BDG2, placing 3rd on accuracy (within **5.3%** of the top result) while running almost **4 times faster** and showing the highest seed-stability among the top three methods.

Comprehensive Survey on Speech Emotion Recognition | Under Review

2026

- International Journal of Speech Technology (Springer Nature).
- Conducted a **PRISMA-based review of 60 papers** spanning traditional ML, deep learning, hybrid/multimodal, and self-supervised approaches (wav2vec 2.0, HuBERT, WavLM) for speech emotion recognition, identifying open gaps in **cross-corpus generalization, explainability, and Green AI deployment**.

ACE-SER: Adaptive, Efficient & Explainable Speech Emotion Recognition | Final-Year Project

2026

- Submitted as **Green SER** to SCOPES 2027 and institutional copyright filed.
- Designed three modules : **ASCA** (Adaptive Speech Complexity Analyzer), **GIE** (Green Inference Engine), and **ASRE** (Audible Saliency Reconstruction Engine) for complexity aware, explainable speech emotion recognition and retained **95.13%** of full-HuBERT accuracy at **52.19%** heavy-branch invocation, with perturbation testing confirming statistically significant explanation faithfulness ($p < 10^{-100}$).

PROFESSIONAL EXPERIENCE

SDE Intern

AyriTech

May 2026 – Present

Hybrid

- Contributing to a multi-modal emotional AI agent product within a 20+ engineer cohort, building RAG-based conversational memory (LangChain, vector database retrieval) and async backend routing (FastAPI, WebSockets, Redis session state).

Software Development Intern

Tata Consulting Engineers Limited (TCE)

Dec 2025 – Jan 2026

Mumbai, India

- Built **SmartCafe**, a full-stack Cafe Management System (React, PostgreSQL) with inventory tracking, order processing, and analytics dashboards for enterprise workflows.

Technical Head – IEEE Student Branch

Vidyalankar Institute of Technology

Sep 2025 – Present

Mumbai, India

- Lead **20+ developers**, organized technical workshops, hackathons, and full-stack/ML mentorship programs and spearheaded migration of the IEEE VIT Mumbai website to React with Supabase backend (GitHub).
- Drove **~35%** increase in event engagement through interactive UI, dynamic event pages, and Git-based agile collaboration across the team.

PROJECTS

FinGuard AI | Multi-Agent Finance Fraud Detection – LangGraph, RAG, FastAPI, React (In Progress)

2026

- Building a multi-agent fraud detection system with RAG pipelines and LangGraph orchestration, LangSmith tracing, RAGAS-based evaluation, and LoRA/QLoRA fine-tuning, served via FastAPI with a React frontend.

Faculty Evaluation App | React.js, Vite, Supabase (PostgreSQL), CSS, Netlify

GitHub | Jun – Sep 2025

- Web application to digitize faculty feedback with React/Vite frontend, Supabase for real-time database and auth, and admin dashboards for insights and deployed on Netlify under principal's mentorship.

TECHNICAL SKILLS

Languages: Python, JavaScript, C++, HTML/CSS

AI/ML & GenAI: LangChain, LangGraph, RAG, Vector Databases, LoRA/QLoRA, PyTorch, TensorFlow, Scikit-learn, BERT/Transformers, wav2vec 2.0, ANFIS

Frameworks & Tools: React.js, Next.js, Vite, FastAPI, Flask, Git & GitHub, Netlify

Databases: PostgreSQL, Supabase, SQL

Core Competencies: Agent Orchestration, ML Research, Full-Stack Development, RESTful APIs & Legal NLP

ACHIEVEMENTS & ADDITIONAL INFORMATION

CDAC – Cloud Computing (Short-Term Course)

Centre for Development of Advanced Computing

- Trained on cloud fundamentals, Linux, and virtualization; hands-on with cloud deployment workflows.

Languages Spoken: English, German, Hindi, Marathi, Kannada